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An Effective Text Steganographic Scheme Based on Multilingual Approach for Secure Data Communication

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***Abstract*—Nowadays, as communication through different internet mediums is increasing rapidly, information security is a huge concern among people of all backgrounds. Text steganography is the method of ensuring data security by concealing any secret message using cover media in such a way that no naked eyes can identify the modification of cover texts. This paper proposes a Bit-Level Embedding Technique, a variant of text steganography, to hide secret messages in the cover medium by manipulating the white spaces available in a text medium and utilizing various Latin alphabets while assuring its confidentiality not attracting suspicion from its appearance. Finally, several experiments and comparisons have been conducted to demonstrate the superiority of the proposed scheme in terms of embedding capacity, invisibility, robustness, and security that make this a standard text steganographic method. The authors propose a data hiding technique where 5-bit can be embedded per embeddable location, which is very rare in text steganography, and the utilization of multilingual forms of English alphabets gives this technique a superior performance.**

***Index Terms*—Data Hiding, Information Security, Secure Communication, Steganography, White-space Characters**

I. Introduction

Today's companies work in a diverse environment where relevant data is extremely valuable. As people use digital devices for online shopping, banking transactions, social media communication so users' data confidentiality has become a major concern for information security [1], [2]. Moreover, any unauthorized access to sensitive and confidential information causes financial and reputational damages, also a threat to national security [3], [4]. One method of obtaining a secure communication medium and protecting data during transmission is to conceal information. Steganography is a process of hiding confidential information in plain text, images, audio, and videos without raising suspicion in unauthorized user's eyes [5]. Fig.1 illustrates the basic function of a text steganographic scheme.

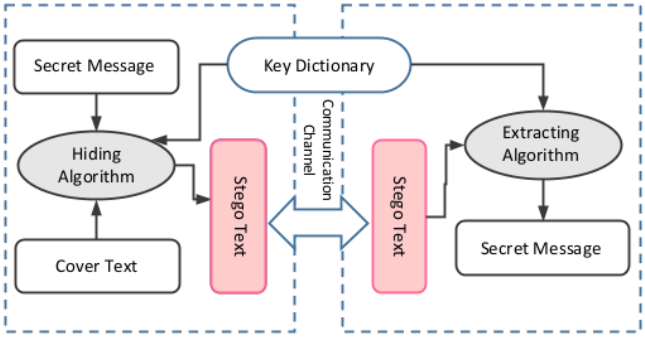

Fig. 1. Block diagram of the text steganography

TABLE I
Description of various symbol

Abbreviations	Descriptions
CT	Cover text
EC	Embedding capacity
LF	Latin form
MF	Multilingual form
SC	Special character
SM	Secret message
ST	Stego text
WSC	White-space character

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TABLE II
LIST OF THE UPPERCASE LETTER AND THE LOWER CASE LETTER USED IN THIS PROPOSED METHOD

Latin	Character	A	C	E	H	I	K	M	N	O	P	S	T	Y
Form	Unicode (U+)	0041	0043	0045	0048	0049	004B	004D	004E	004F	0050	0053	0054	0059
Multilingual	Character	A	C	E	H	I	K	M	N	O	P	S	T	Y
Form	Unicode (U+)	0391	0421	0415	041D	0399	039A	039C	039D	041E	0420	0405	0422	04AE
Latin	Character	a	c	d	e	g	i	j	l	m	o	p	s	y
Form	Unicode (U+)	0061	0063	0064	0065	0067	0069	006A	006C	006D	006F	0070	0073	0079
Multilingual	Character	a	c	d	e	g	i	j	l	m	o	p	s	y
Form	Unicode (U+)	0430	0441	0501	0435	1D5C0	0456	03F3	04CF	217F	043E	0440	0455	0443

Table.I illustrates the description of various symbols used in this paper. Steganography is often confused with cryptography because both process are used to secure sensitive and important information. But, a cryptographic object raises suspicion among third parties, while in case of steganographic object none does not even notice the existence of any secret message [6].

The major drawback of cryptography is that encrypted data is not hidden, and one can understand the presence of valuable information from the encrypted data. So, if given enough time, there is a possibility that a third party could eventually decrypt the encrypted data [7]. Steganography can help to solve this major problem in cryptography. Steganography is the procedure of concealing confidential information in innocent-looking text, images, audios, and videos. The use of steganography protects data from being detected in the first place [8].

Text steganography aims to establish a stable interaction that is unassuming and avoids drawing suspicion to transmit confidential data [9]. In-text steganography, the secret message (SM) is embedded on a cover text (CT) which is actually a plain text. When the SM is embedded on the CT, then the embedded text is called stego text (ST). Furthermore, the approved user receives the stego text through a medium. To ensure the security of the stego text, a stego key is required. Using the stego key, only the intended recipient will retrieve the hidden message from the stego text [9], [10]. The English alphabet has a few distinct features that distinguish it from different alphabets. One of the features is called multilingual form. Every letters in the English alphabet have multiple different Unicode of different languages. These letters are called multilingual characters [5]. The appearance of white-space characters (WSC) is a way to conceal confidential messages into a text document. A WSC does not have any visible sign but does consume space. The security and embedding capacity (EC) of any text steganographic scheme can be immensely increased by using the WSC [10]–[13].

In this paper, the authors propose an effective data hiding method to ensure secure communication using the multilingual form of English alphabets and multiple WSC. A bit-level embedding procedure has been introduced for hiding the SM efficiently into an innocent-looking text file. This steganographic scheme utilizes inter-word spacing to hide the secret message. Here, the authors use a different combination of white spaces, various English letters having Latin form and multilingual form shown in Table. II for hiding the binary bit stream converted from the actual SM, so it is completely invisible and yields nothing to a human investigation. Besides, this approach can embed five bits per embeddable location, making this scheme superior to most other existing methods. The embedding and extracting procedure of proposed scheme is unique and dictionary-based. To begin with, the interceptor will face difficulty determining whether it is a ST or not. Because the selected multilingual characters are similar to the normal Latin alphabets shown in Table.II, the interceptor must determine which and where the encrypted character is placed. Finally, even if someone discovers the existence of SM, the list of special characters and the binary bit stream combination are required to extract the SM. So, it secures these sensitive data more than ever before. Moreover, the authors consider invisibility, security, robustness, and embedding capacity parameters while creating this text steganographic scheme [14].

This paper proposes an effective text steganography approach that achieves the ideal combination of embedding capacity, confidentiality, robustness, and invisibility. The proof of the superiority of the proposed method rather than the existing methods is being discussed in

TABLE III
LIST OF USED WHITE SPACE CHARACTERS AND IT'S CORRESPONDING HIDDEN BIT

Used Space	Unicode (U + '4 digit')	Total Width	Hidden bit (Char.in LF)	Hidden bit (Char.in MF)
3 Per em + Mongolian Vowel Separator	2004 + 180E	1/3 em	00000	10000
Thin Space + Mongolian Vowel Separator	2009 + 180E	1/4 em	00001	10001
Punctuation Space + Mongolian Vowel Separator	2008 + 180E	1/3 em	00010	10010
Medium Mathematical + Mongolian Vowel Separator	205F + 180E	1/4 em	00011	10011
3 Per em + Word Joiner	2004 + 2060	1/3 em	00100	10100
Thin Space + Word Joiner	2009 + 2060	1/4 em	00101	10101
Punctuation Space + Word Joiner	2008 + 2060	1/3 em	00110	10110
Medium Mathematical + Word Joiner	205F + 2060	1/4 em	00111	10111
3 Per em + Mongolian Vowel Sep. + Word Joiner	2004 + 180E + 2060	1//3 em	01000	11000
Thin Space + Mongolian Vowel Sep. + Word Joiner	2009 + 180E + 2060	1/4 em	01001	11001
Punctuation Space + Mongolian Vowel Sep. +Word Joiner	2008 + 180E + 2060	1/3 em	01010	11010
Medium Math. + Mongolian Vowel Sep. + Word Joiner	205F + 180E + 2060	1/4 em	01011	11011
3 Per em + Word Joiner + Mongolian Vowel Sep.	2004 + 2060 + 180E	1/3 em	01100	11100
Thin Space + Word Joiner + Mongolian Vowel Sep.	2009 + 2060 + 180E	1/4 em	01101	11101
Punctuation Space + Word Joiner + Mongolian Vowel Sep.	2008 + 2060 + 180E	1/3 em	01110	11110
Medium Math. + Word Joiner + Mongolian Vowel Sep.	205F + 2060 + 180E	1/4 em	01111	11111
Punctuation Space + Word Joiner + Word Joiner	2008 + 2060 + 2060	1/3 em	0	1

the result and comparison section of this paper.

The remained portions of the paper are arranged as follows. Section II discusses the related works in this field, Section III discusses the methodology and Section IV discusses the experimental results and the evaluation of the proposed work compared to the other works. Finally, in Section V, the future scope is discussed and concludes this paper.

II. RELATED WORKS

This section discusses about some of the significant and similar text steganographic methods to the proposed scheme.

A. Bit-level Embedding Techniques

1) Unicode Space Characters: There are around 30 Unicode space characters besides the normal space known as white space characters (WSC). Some of these WSC is look-alike the normal space. Some data hiding method in text steganography was proposed by taking advantage of WSC.

The very first data hiding method based on WSC was proposed by Kwan [15]. It is a steganographic tool which is known as SNOW. Here, the authors embed SM into cover document by giving extra WSC at the end of line. McKellar proposed a data hiding tool named Spammimic [16]. In this scheme, they hide binary bits by using end of line spacing and inter paragraph spacing. Por et al. proposed a data hiding method using inter word spacing and inter paragraph spacing [10]. This steganographic method is also known as whitesteg.

M. Shazzad-Ur-Rahman et al. proposed an efficient text steganography using Bengali text as cover object. In this method, a unique data hiding method is proposed using the duality form of Bengali character and WSC [17].

Por et al. proposed a WSC based data hiding method where they hides almost two binary bits in every WSC. This method is also known as UniSpaCh [13].

In this approach, secret binary bits are embedded by inter-sentence, inter-word, inter-paragraph and end-of-line spacing which help to improve the EC than before.

Kumar et al. proposed another method like UniSpaCh using the advantage of total number of 9 WSC [12]. The usage of 9 WSC has increased the embedding capacity of this scheme, but the lack of security and imperceptibility occurred because of the normalization of the font size.

Singh et al. proposed a text steganographic scheme by adding extra null spaces to hide data. In this method, to embed few binary bits it takes more space, which is a big disadvantage of this method [18].

2) Line and Word Shifting Method: Several researchers proposed a data hiding technique based on line shifting and word shifting method in [19]. Here, the secret bit is embedded by shifting the lines of the text some degrees vertically. In the word shifting method, secret data is embedded by moving the middle word and keeping the adjacent words, particularly in a line [20]. The security of the hidden message can be destroyed using OCR programs and retyping text.

3) Linguistic Approach: Most of the English letters can be written in another Unicode besides its natural Latin Unicode. A unique text steganographic scheme was discussed by using the multilingual form of several English letters. In this method, '2' binary bits was embedded in per embeddable location [5]. A duality form based text steganographic method was proposed in [21]. Here, SM was hidden into cover object by taking the advantage of classical and non-classical form of some Bengali letters.

4) Change Tracking Approach: Liu et al. proposed a text steganographic scheme in Microsoft Word document where secret bits are embedded in a CT by changing track [22].

B. Character Level Embedding Methods

1) Using Font Attribute: A novel data hiding method was proposed using property of similar English font types [23]. In this techniques SM is embedded only in the upper case letter of the CT. The visual similarity of different fonts is the main carrier of the system.

In text steganography, some methods are developed based on marking the characters of CT document. In this approach, some methods are developed by changing height of character [24], normalizing font size [25]. Here, secret characters are embedded by making text bold or giving underline. A novel technique was proposed in [26] using font attribute called character spacing to hide payload into text document. Here, two operations; Frequency Normalization Set (FNS) in combination with the Character and String Mapping (CSM) was introduced to make the method more efficient. Agarwal et al. proposed several data hiding methods by generating a list of words, missing letter puzzle and also hiding data in paragraphs [25].

III. Proposed Work

In this paper, a unique text steganographic scheme is proposed to ensure secure hidden communication. This section discusses the embedding and extracting procedure of this proposed text steganographic scheme. Here, Fig.2 represents the data hiding process of the proposed scheme and Algorithm.1 describes the extracting method.

A. Data Extracting Algorithm

Algorithm 1: Extracting Procedure

```
Result:
Secret message in bits form
Secret message in text form
Input: Stego document with the secret message;
List with natural latin form, List1 = ['A', 'C',.,.,]
List with multilingual form, List2 = ['A', 'C',.,., ]
while character in stego text do
    if white-space.found() then
        for i = Index of(special-space - 1), i
          >position of previous special space,
          i=i+1 do
            if multilingual-character.found() then
                result = result + corresponding
                  bits of the special characters
                  based on the dictionary.
            else
                continue
            end
        end
    else
    end
end
```

B. Data Hiding Method

Fig.2 illustrates the process of hiding secret message into the cover document.

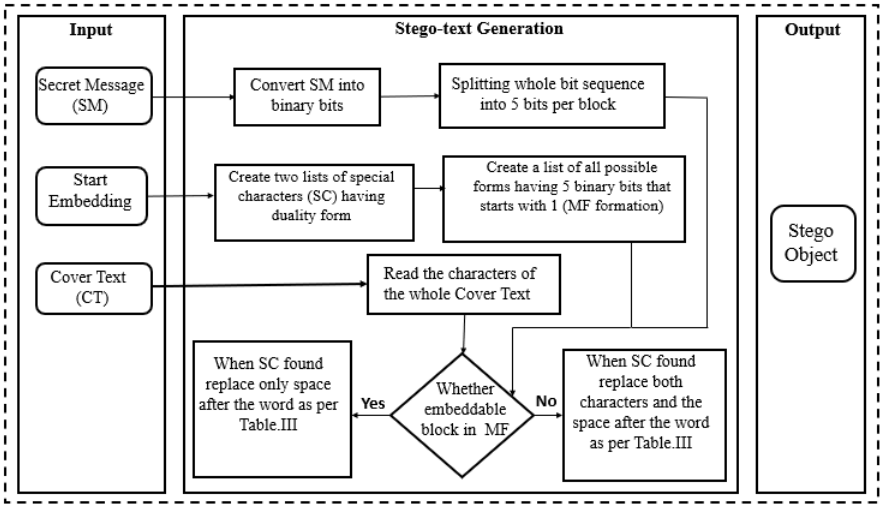

Fig. 2. Embedding procedure of our proposed scheme.

In this method, the advantage of multilingual form of English letters was utilized to embed binary bits into the cover text. All the English alphabet has more than 3 multilingual form and in some cases both the Latin form (LF) and the multilingual form (MF) look like the same. The selected special characters (SC) having multilingual form which is used in this scheme is shown in Table. II.

Moreover, the usage of WSC improves the security and the EC of this text steganographic scheme. In this scheme, total 6 white-space characters are used to hide the SM and these WSC are quite similar to the normal space. Table.III represents the combination of WSC and it's corresponding hidden bits.

IV. Experimental Results and Comparisons

This section evaluates the efficiency of this method and some comparison between the existing techniques will be illustrated here. This data hiding method is implemented in Visual Studio Code 2021, which provided the entire environmental features. To develop the text steganographic scheme Python programming language (version 3.7) is used. Fig.3 illustrates the GUI of our proposed scheme.

A. Evaluation Criteria Analysis

An ideal text steganographic scheme has to have a better rate of embedding capacity, higher imperceptibility, robustness and security.

1) Embedding Capacity (EC): The cumulative number of secret bits that can be concealed on the cover document is called the embedding capacity (EC). This steganographic scheme has a higher rate of EC as it can embed 5-bit per embeddable location.

$$EC = \frac{\text{bits of secret text}}{\text{bits of cover text}}$$

Fig.5 illustrates the EC comparison between some existing methods. The model with the highest embedding capacity uses smaller cover documents to embed the same amount of payload, hence this proposed scheme has a higher embedding capability than current models. In text steganography, secret message is also called as payload.

2) Imperceptibility: Imperceptibility is one of the most important characteristics of any steganographic model. When there is no visual difference between the innocent-looking cover text and the secret message that contained stego text, that characteristic is called imperceptibility if any steganographic model lacks the invisibility of stego-object and creates suspicion security breach. This model has a better rate of imperceptibility as both CT and the ST look the same. Fig.4 illustrates the invisibility of the proposed scheme.

3) Robustness: Robustness means the capability of coping up with a variety of environments and can fix the error itself. An ideal ST always maintain standard and does not become vulnerable in various platforms. This method has a higher robustness as it has higher

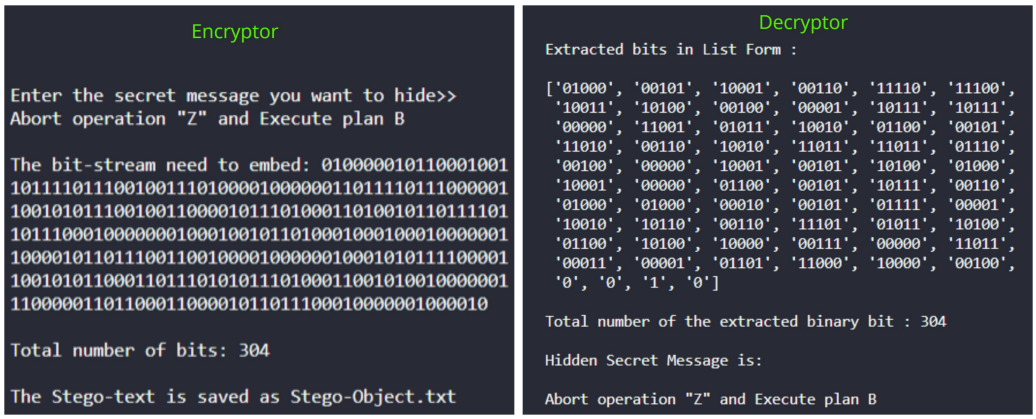

Fig. 3. Embedding and extracting GUI of the proposed scheme.

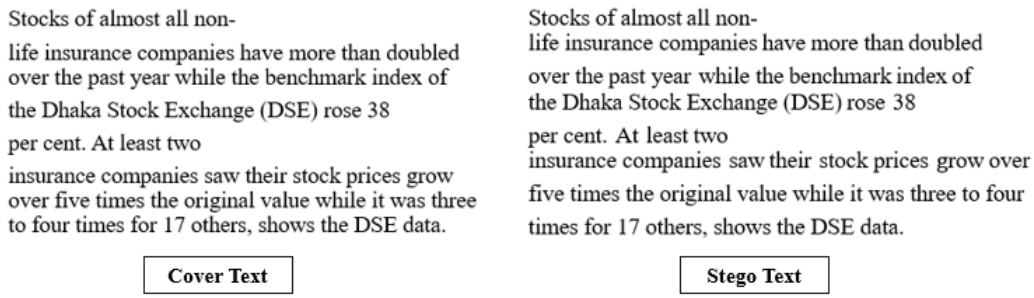

Fig. 4. Invisibility of the proposed scheme

TABLE IV
COMPARISON OF THE PROPOSED METHOD WITH THE EXISTING TECHNIQUES

Technique	Type of embedding	Level of stealthiness (based on visibility)	No. of distortions per secret character	Embedding probability	Possibility of non-occurrence of alphabets is handled?	No. of embeddable locations per secret character	No. of cover characters per secret character (approximate)
Hiding data in wordlist [25]	Character-level	Low	1	Uniform	Yes	1	10.5
Inter-sentence spacing [10]–[13]	Bit-level	Low	8	Uniform	Yes	8	751
Line shifting [19]	Bit-level	Low	8	Uniform	Yes	8	1020
Word shifting [20]	Bit-level	High	8	Uniform	Yes	8	98.5
UniSpaCh [13]	Bit-level (2-bits)	High	4	Uniform	Yes	4	26.5
Font attribute [26]	Character-level	High	1	Close to uniform	Yes	4	4
Proposed method	Bit-level	High	3	Uniform	Yes	2	12

rate of distortion robustness in social media application like WhatsApp, Twitter, Messenger and SMS service in android phone which is shown in Table.V. It is a very important criteria of any text steganographic scheme. The security of any scheme highly depends on it's robustness. Table.V shows various platform where the stego text of the proposed method does not become vulnerable.

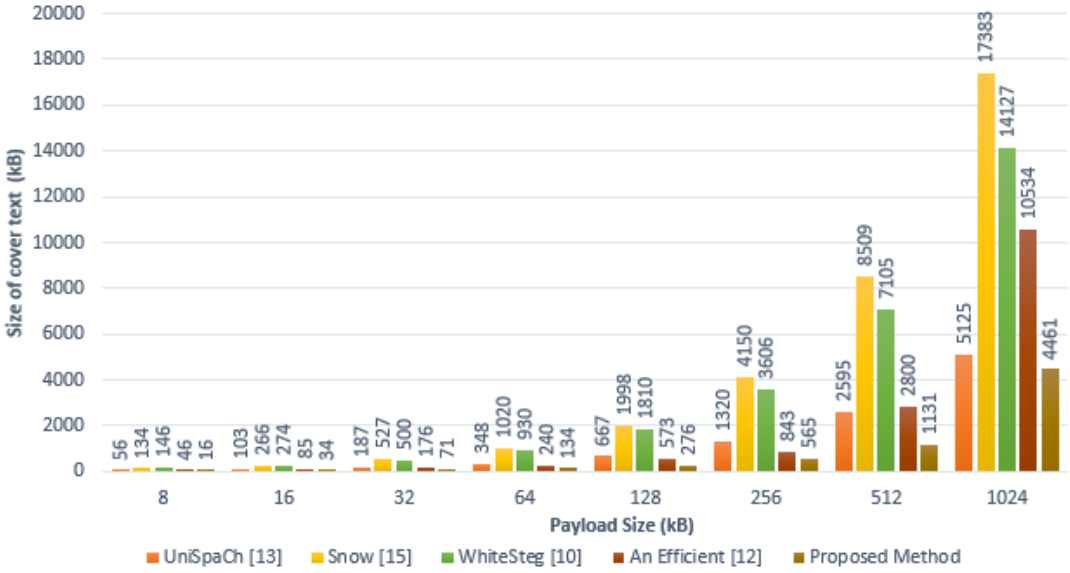

Fig. 5. EC comparison between some existing methods

TABLE V
INVISIBILITY AND ROBUSTNESS ANALYSIS IN VARIOUS PLATFORMS

Online Platform	Invisible	Robust
Facebook	Yes	No
Messenger	Yes	Yes
WhatsApp	Yes	Yes
Twitter	Yes	Yes
SMS (Android)	Yes	Yes
Gmail	Yes	Yes
Viber	Yes	Yes

4) Security: Security of a steganographic technique means to ensure the confidentiality of the secret message from unauthorized users. This method can secure the secret message because it achieves the optimally possible balance of three other criteria: higher embedding capacity, high imperceptibility, and high robustness. It can also ensure the confidentiality of this data hiding method because of the usage of the WSC that are roughly the same length as normal space. So, it has become almost impossible for a third party to retrieve the secret message from the ST. The huge amount of multilingual forms of the Latin alphabets gives this technique extra security. Moreover, the imperceptibility of the stego document is very high, which keeps the secret message invisible to the third party.

V. CONCLUSION AND FUTURE WORK

This paper has introduced an effective Bit-Level embedding text steganographic algorithm where 5-bit of data can be stored per embeddable location, which is unique on its own. This 5-bit embedding capacity combined with a vast array of usable characters to choose the location from has given this method an edge over previous works in this sector. This method mainly manipulates the white spaces and certain English alphabets, maintaining its secrecy quite well and not attracting suspicion in its visual appearance.In addition, as bit-level data is embedded in the plain text file, it has various interesting usage and applications. This method is also suitable for embedding audio, video, or image file. Besides, this proposed method can have a major impact in securing used data in online communication and transactions by combining it with cryptography.

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